VIETNAM NATIONAL UNIVERSITY – HO CHI MINH CITY

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CLASS: CC02

COURSE PROJECT REPORT

Project General Topic: Basics of and Applications of Solar Energy Physics and Engineering

Project Specific Topic: Solar Irradiance Analysis for Photovoltaic Systems in Ho Chi Minh City

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ABSTRACT

Vietnam's installed photovoltaic capacity reached 18,667 MW by the end of 2024, making optimal panel configuration increasingly critical for energy and economic outcomes. For fixed-tilt rooftop systems — the most widely adopted configuration — tilt angle and azimuth orientation govern the magnitude of incident solar irradiance through fundamental solar geometry principles, including the declination angle, hour angle, and angle of incidence on a tilted surface. A south-facing orientation maximizes annual irradiance capture, yet concentrates generation during midday hours when electricity prices under Vietnam Electricity's (EVN) industrial time-of-use tariff are at their lowest. This study quantifies this physical–economic trade-off for Ho Chi Minh City (10.77°N, 5.09 kWh/m²/day) using a solar geometry irradiance model with temperature-corrected PV efficiency (NOCT model), implemented in MATLAB across tilt angles of 0°–60° for south and west orientations. Results show that the physically optimal configuration (β = 15°, south, 788.59 kWh/year) differs from the economically optimal configuration (β = 20°, west, 1,577,373 VND/year), with the latter achieving 0.56% higher annual revenue at a cost of 2.60% reduction in total energy output. These findings offer direct guidance for homeowners and small businesses evaluating rooftop PV installations in tropical urban environments.

1. INTRODUCTION

1.1. Background of Solar Energy

- Vietnam has emerged as one of Southeast Asia's leading solar markets, reaching a cumulative installed photovoltaic (PV) capacity of 18,667 MW, according to the International Renewable Energy Agency. Ho Chi Minh City is particularly well-positioned for solar deployment: the city receives an average annual irradiance of approximately 1,581 kWh/m²/year, with a daily average global horizontal irradiance (GHI) of 5.09 kWh/m²/day — the highest among Vietnam's major urban centers. This high and seasonally stable solar resource makes rooftop PV systems an increasingly attractive investment for residential and commercial consumers in the city.
- Among available configurations, fixed-tilt rooftop systems are the most widely adopted due to their low capital cost and minimal maintenance requirements. However, as established in the solar engineering literature, the tilt angle and azimuth orientation of a fixed panel directly govern the magnitude of incident irradiance and, consequently, system energy yield.
- The physical behavior of a fixed-tilt PV system is governed by classical solar geometry principles — specifically, the declination angle, hour angle, and the resulting angle of incidence on a tilted surface — which together determine the instantaneous irradiance available for conversion. Combined with a temperature-dependent efficiency model accounting for real operating conditions, these principles form the physical foundation of the present analysis.

1.2. Problem Statement

- A fundamental tension exists between physical and economic optimization of fixed-tilt PV systems. South-facing panels at near-latitude tilt angles maximize annual energy yield, yet concentrate generation during midday hours — a period corresponding to the lowest electricity price tier under Vietnam Electricity's (EVN) industrial time-of-use (ToU) tariff. Conversely, westward orientation shifts peak generation toward the late afternoon (17:30–22:30), which coincides with EVN's peak pricing window, where rates reach up to 3,640 VND/kWh — approximately 2.8 times the off-peak rate of 1,300 VND/kWh. This creates a scenario in which a westward configuration may achieve lower energy output yet higher annual revenue. Despite the practical relevance of this trade-off for PV adopters in Ho Chi Minh City, it has received limited systematic quantitative treatment in the local context.
- The practical implications of this trade-off extend directly to homeowners, small businesses, and urban developers in Ho Chi Minh City who must decide on panel orientation at the time of installation — a decision that, once made, determines system economics for the 20–25 year panel lifetime.

1.3. Objectives of the Study

This study addresses the above gap through the following objectives:

1. To model incident solar irradiance on a tilted PV surface across a range of tilt angles (0°–60°) and azimuth orientations at the latitude of Ho Chi Minh City (10.77°N).
2. To estimate hourly and annual electrical energy output using a temperature-corrected efficiency model.
3. To calculate annual revenue for each configuration under EVN's industrial ToU tariff and identify the economically optimal tilt angle and orientation.
4. To quantify the trade-off between the physically optimal and economically optimal configurations and discuss implications for practical system design.

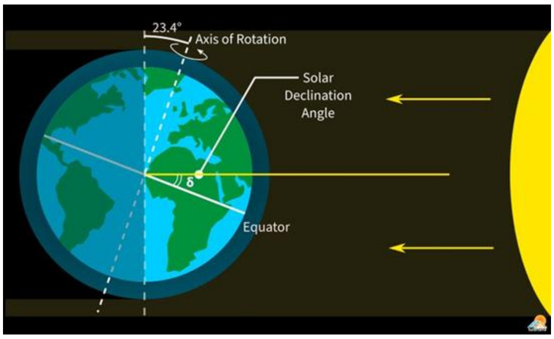
2. BASIC PHYSICS

2.1. Solar Geometry

- **The declination δ:**
  - The angular position of the sun at solar noon (i.e., when the sun is on the local meridian) with respect to the plane of the equator, north positive; $-23.45^\circ \leq \delta \leq 23.45^\circ$
  - The declination δ can be found from the approximate equation of Cooper (1969)

$$\delta = 23.45 \times \sin\left(360 \times \frac{284+n}{365}\right)$$

(Where n is the day of the year, e.g., n=1 for January 1).



- **Hour angle ω**: The angular displacement of the sun east or west of the local meridian due to rotation of the earth on its axis at 15° per hour; morning negative, afternoon positive (equals 0 at 12 p.m)

$$\omega = (LST - 12) \times 15^\circ$$

- **Slope β:** The angle between the plane of the surface in question and the horizontal; $0^\circ \leq \beta \leq 180^\circ$ ($\beta > 90^\circ$ means that the surface has a downward-facing component.)
- **Latitude φ:** The angular location north or south of the equator, north positive; $-90^\circ \leq \varphi \leq 90^\circ$
- **Surface azimuth angle γ:** The deviation of the projection on a horizontal plane of the normal to the surface from the local meridian, with zero due south, east negative, and west positive; $-180^\circ \leq \gamma \leq 180^\circ$
- **Zenith angle θ_z**: The angle between the vertical and the line to the sun, that is, the angle of incidence of beam radiation on a horizontal surface.

$$\cos\theta_z = \cos\varphi\cos\delta\cos\omega + \sin\varphi\sin\delta \text{ (with } \varphi = 10.77^\circ \text{ HCM city's latitude)}$$

- **Angle of incidence θ**: The angle between the beam radiation on a surface and the normal to that surface, depending on slope β and surface azimuth angle γ

$$\cos\theta = \sin\delta\sin\varphi\cos\beta - \sin\delta\cos\varphi\sin\beta\cos\gamma + \cos\delta\cos\varphi\cos\beta\cos\omega + \cos\delta\sin\varphi\sin\beta\cos\gamma\cos\omega + \cos\delta\sin\beta\sin\gamma\sin\omega$$

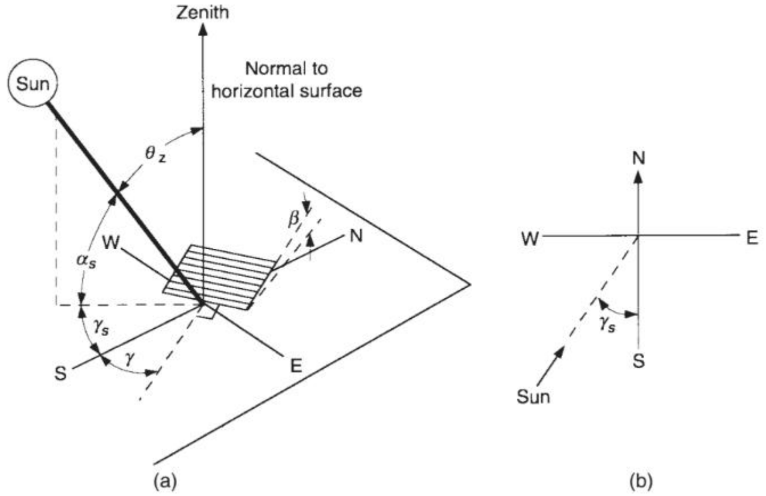


Figure 1.6.1 (a) Zenith angle, slope, surface azimuth angle, and solar azimuth angle for a tilted surface. (b) Plan view showing solar azimuth angle.

2.2. Solar Irradiance Models

- Extraterrestrial radiation G_{0n}: $G_{0n} = G_{sc} \times [1 + 0.033 \times cos(\frac{360n}{365})]$ with G_{sc} = 1367 W/m²

  G_{0n} : the extraterrestrial radiation incident on the plane normal to the radiation on the nth day of the year

- Solar radiation on a tilted surface G_T: $G_T = G_{0n} \times k_T \times cos(\theta)$, where k_T is the clearness index, using the monthly average values for Ho Chi Minh City obtained from available reference tables.

- $G_T = G_{sc} \times [1 + 0.033 \times cos(\frac{360n}{365})] \times k_T \times cos(\theta)$

  Equations:

  G_{0n}=$G_{sc} \times [1 + 0.033 \times cos(\frac{360n}{365})]$

  $G = k_T . G_{0n}$

  G_T=$G . cos(\theta)$

  G_T=$k_T . G_{0n} . cos(\theta)$

- The radiation on a tilted surface is proportional to $cos(\theta)$, where θ is the angle of incidence between the solar beam and the normal to the panel surface.
- When $cos(\theta)$ increases, the sunlight becomes more perpendicular to the solar panel, allowing more solar energy to be concentrated on the surface.
- Therefore, a larger value of $cos(\theta)$ results in stronger received solar radiation and higher energy absorption by the panel.

2.3. Photovoltaic Physics and Power Output

2.3.1 Quantum Nature of Light and Photoelectric Effect

- According to Planck's quantum theory, a light beam is a stream of energy particles called photons. When a photon interacts with a metal surface, it collides with an electron. In this interaction, the photon transfers its entire energy to the electron.
- The energy of a photon is determined by Einstein's formula:

$$E = h \times f$$

h: Planck constant (6.626×10^{-34} J.s)

f: Incident light frequency (Hz)

- Photon energy absorption: During the interaction between a photon and an electron, the electron cannot absorb only a portion of the energy and let the rest pass through; it either accepts all of it or nothing at all. This energy must overcome a specific threshold of the material to cause an effect.
  - In Metals, this threshold is the Work Function (A).
  - In semiconductors, this threshold is the Bandgap (E_g).

Comparison Feature	Case 1: h: f < A	Case 2: h: f ≥ A
Status	FAILURE	SUCCESS
Mechanism	Electrons receive energy but not enough to overcome the binding force (Work function A).	Photon energy is greater than or equal to the Work function (A) of the metal.
Result	Energy is converted only into thermal energy,	Electrons overcome the binding force and are

	heating the metal.	ejected from the metal surface.
Phenomenon	No electrons are released.	Excess energy is converted into maximum kinetic energy of the emitted electron.

Table 4. *Photoelectric Effect in Metals: Case Comparison*

- Interaction with Silicon Atoms:
  - When a photon with energy $E = h \times f$ collides with a silicon atom, if this energy is greater than the Bandgap (Eg), it will excite an electron from the valence band to the conduction band, creating an electron-hole pair ($e^- - h^+$).
  - In solid-state physics, electrons are concentrated in two bands:
    - Valence Band: Where electrons are tightly bound.
    - Conduction Band: Where electrons move freely to generate current.
  - The energy gap between these two bands is the Bandgap (Eg).

Comparison Feature	**Case 1: h: f < E_g**	**Case 2: h: f ≥ E_g**
Status	FAILURE	SUCCESS
Mechanism	Photon energy is not enough to bridge the **Bandgap (E_g)**.	Photon energy exceeds the **Bandgap (E_g)**.
Result	Passes through or generates heat.	Electron is excited about the **Conduction Band**.
Phenomenon	No current generated.	Creation of **electron-hole pairs** to generate electricity.

Table 5. *Photovoltaic Effect in Silicon: Case Comparison*

2.3.2. Efficiency Limits (Shockley-Queisser Limit)

- Silicon cell efficiency is theoretically capped at ~33.7% (practically 15–22% in industry) due to three main losses:
  - Non-absorption: $E_{photon} < E_g$
  - Thermalization: $E_{photon} > E_g$
  - Blackbody Radiation**:** Any object above absolute zero radiates energy; solar cells also re-radiate energy back into the environment**.**

Stage	Remaining Efficiency	Main Cause	Evidence
Solar radiation	100%	Initial energy	Solar Constant
Physical limits	~33.7%	Photons too weak or too strong	Shockley & Queisser (1961)
PV cell (Lab)	~26%	Material optimization	Martin Green (UNSW records)
Industrial Production	15% - 22%	Production costs, resistance, dust, actual temperature	Standard Test Conditions (STC)

Table 6. *Shockley-Queisser Efficiency Loss Stages*

2.3.3. Optical Losses (Fresnel Reflection)

- When sunlight passes through the interface of environments with different refractive indices (air and the glass protective layer), part of the radiation cannot enter the semiconductor cell.
- Fresnel's Law of Reflection explains that when the angle of incidence θ is large (very slanted light), the reflection coefficient increases significantly.
- Consequence: A large portion of radiation is reflected back into space instead of penetrating the glass to reach the cell surface.
- Significance: This helps engineers calculate the optimal tilt angle of the panel to maximize the amount of photons entering the semiconductor layer.

2.3.4. Temperature Model and Actual Efficiency

- Determine the actual operating temperature of the photovoltaic cell based on environmental conditions:

$$T_C = T_a + \left(\frac{NOCT - 20}{800}\right) \times G_T$$

T_c: Actual temperature (°C)

T_a: Ambient temperature (°C)

G_T: Solar radiation (W/m^2)

NOCT: Nominal Operating Cell Temperature, usually provided by the manufacturer (42°C).

- The relationship between temperature and efficiency is expressed via the formula:

$$\eta = \eta_{ref} \times [1 - \beta_T \times (T_C - T_{ref})]$$

η: Efficiency of the PV cell under actual conditions.

η_{ref} : Efficiency at Standard Test Conditions (STC).

β_T : Temperature degradation coefficient (material property).

T_C : Actual temperature of the PV cell.

T_{ref} : Standard reference temperature.

- Operational Paradox: While stronger solar radiation produces more current, it also causes the panel to heat up rapidly. When the cell temperature (T_C) rises high above the reference level, it reduces the overall photovoltaic efficiency of the system.

2.3.5. Power Output and Energy Yield Formulae

- The formula for the actual power output of the panel (P_{out}) after subtracting the energy reflected back due to the Fresnel effect and the efficiency degradation caused by the heating of the panel.

$$P = G_T \times A \times \eta$$

A: Solar panel area (m^2)

η: Panel efficiency (typically 15-22% for common modern panels)

- Hourly Energy Output Formula:

$$E(t) = P(t) \times \Delta t \ = G_T \ \times A \ \times \eta \times \ 1h \ (kWh)$$

2.4. Time-of-Use Pricing Mechanism

Based on Decision No. 1279/QD-BCT, EVN applies a tariff structure divided into three distinct price brackets to regulate the power grid load. Understanding these specific figures is essential for optimizing the financial return of solar energy systems.

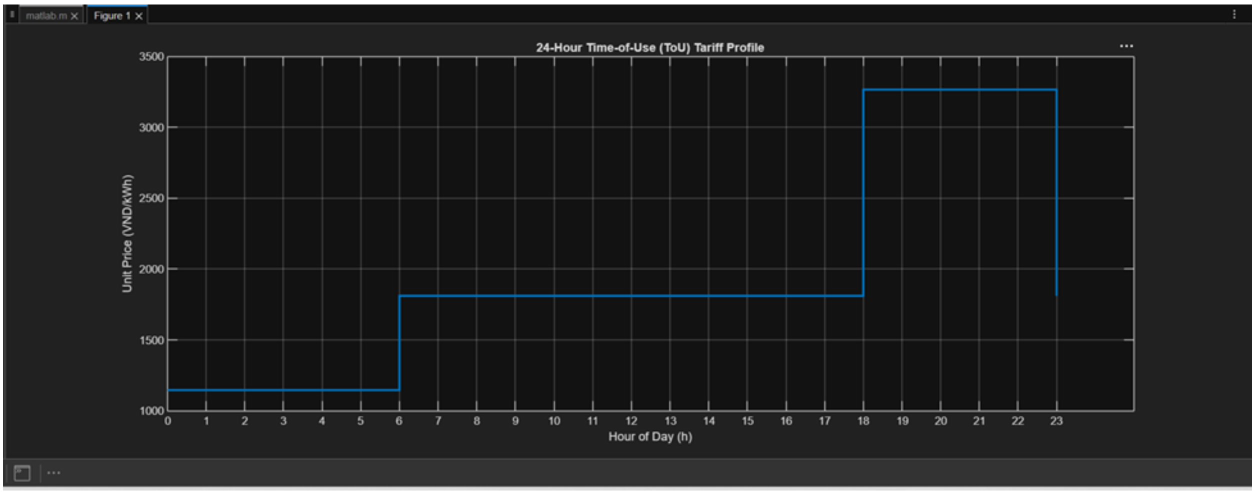
2.4.1. Applicable Time Windows

- Normal Hours: Monday to Saturday (06:00 – 17:30 and 22:30 – 24:00); Sunday (06:00 – 24:00).
- Peak Hours: Monday to Saturday (17:30 – 22:30); No peak hours on Sunday.
- Off-Peak Hours: All days of the week from 00:00 – 06:00.

2.4.2. Electricity Data By Voltage Level (unit VND/kWh)

No.	Customer Group	Electricity Selling Price (VND/kWh)
1	Retail electricity price for manufacturing sectors	
1.4	**Voltage below 6kV**	
	a) Normal hours: 06:00–17:30 & 22:30–24:00 (weekdays + all day Sunday)	1.987
	b) Off-peak hours: 00:00–06:00 (daily)	1.300
	c) Peak hours: 17:30 - 22:30 (Monday-Saturday)	3.640

2.4.3. Revenue Optimization Strategy



- **Tariff Gap:** Peak hour electricity (from 17:30) is priced ~80% higher than Normal hour electricity (midday).
- **Power Output Mismatch:**
  - South-facing: Reaches peak output during 12:00–14:00 but stops generating just as the most expensive price bracket (17:30) begins.
  - West-facing: Shifts the peak generation toward the afternoon, maintaining high output during the high-priced peak window (17:30–20:00).
- Total Revenue

$$R = \sum_{t=1}^{24} (Et \times Pt)$$

R: Total daily revenue (VND).

t: The t^{th} hour of the day (t = 1, 2, ..., 24).

E_t: Energy yield generated during hour t (kWh).

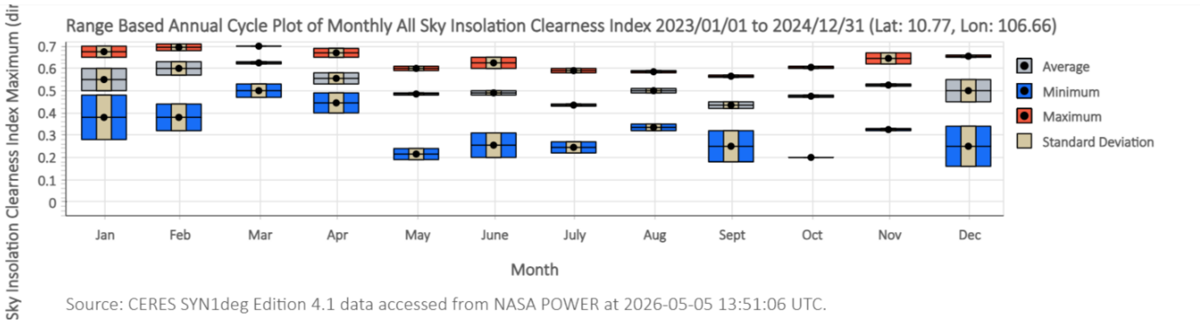
P_t: Electricity unit price at hour t

- Conclusion: Although West-facing systems generate a lower total energy yield (kWh), their higher value per kWh maximizes the Total Revenue.

3. METHODOLOGY AND SIMULATION SETUP

3.1. Local Data

- **Geographic Parameters (Ho Chi Minh City):**
  - **Latitude:** 10.770°N
  - **Longitude:** 106.660°E



Month	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**
k_T	0.55	0.60	0.62	0.55	0.48	0.49	0.43	0.50	0.43	0.47	0.52	0.50

Table 1. *Monthly Clearness Index k_T for Ho Chi Minh City (2023–2024)*

Month	Jan	Feb	March	April	May	June	July	Aug	Sep	Oct	Nov	Dec
Average temperature (°C)	26.54	28.14	29.99	31.10	29.93	27.66	26.90	26.91	26.82	26.70	26.70	26.00

Table 2. *Monthly Average Ambient Temperature T_a for Ho Chi Minh City*

- **Solar Panel Specifications:**
  - Model: Canadian Solar HiKu CS3W-450MS
  - Maximum Power (P_{max}) 450W
  - Panel Area (A): 2.21(2.108×1.048) m^2
  - Temperature Coefficient of Pmax (β_T): -0.35%/∘C
  - Reference Efficiency (η_{ref}) : 20.4%
  - Nominal Operating Cell Temp. NOCT: 42 ± 3∘C
  - Standard Ambient Temp (T_a): 20 (when measuring NOCT) / 25 (STC) ∘C
- **EVN Electricity Tariffs:**

No.	Customer Group	Electricity Selling Price (VND/kWh)
1	Retail electricity price for manufacturing sectors	
1.4	**Voltage below 6kV**	
	a) Normal hours: 06:00–17:30 & 22:30–24:00 (weekdays + all day Sunday)	1.987
	b) Off-peak hours: 00:00–06:00 (daily)	1.300
	c) Peak hours: 17:30 - 22:30 (Monday-Saturday)	3.640

Table 3. *EVN Industrial ToU Electricity Tariff (Voltage below 6 kV)*

3.2. MATLAB Algorithm Implementation

- **Step 1: Finding Solar Angle**
  - Calculate solar parameter: declination angle δ, hour angle ω, and zenith angle θ_Z..
  - Evaluate the incidence angle θ relative to solar panel tilt angle (β) and azimuth angle (γ).
- **Step 2: Solar Radiation Estimation**

- Formula: $G_T = G_o \times k_t(month) \times max(cos(\theta),0)$
- Using $max(cos(\theta),0)$ to filter out negative values when the sun is below the horizon.

- **Step 3: Evaluating Surface Temperature and Actual Efficiency**
    - Surface temperature: $T_c = T_a + (\frac{NOCT - 20}{800}) \times G_T$
    - Actual efficiency: $\eta = \eta_{ref} \times [1 - \beta_T \times (T_c - T_{ref})]$ (with $T_{ref} = 25$ under STC).
- **Step 4: Power and Energy Generated Per Hour**
    - Power formula: $P = G_T \times A \times \eta$
    - Energy generated formula: $E = P \times \Delta t$
- **Step 5: The Revenue Calculation**
    - Formula: $Revenue = \Sigma(E(t) \times Price(t))$
    - Accumulate the revenue over the course of 365 days to get the total revenue.
- **Step 6: Determine The Optimal Tilt Angle Loop**
    - Repeat tilt angle β from 0° to 60°.
    - Save total energy and total revenue of β inside an array.
    - Apply *max(β)* to find the most optimal β.

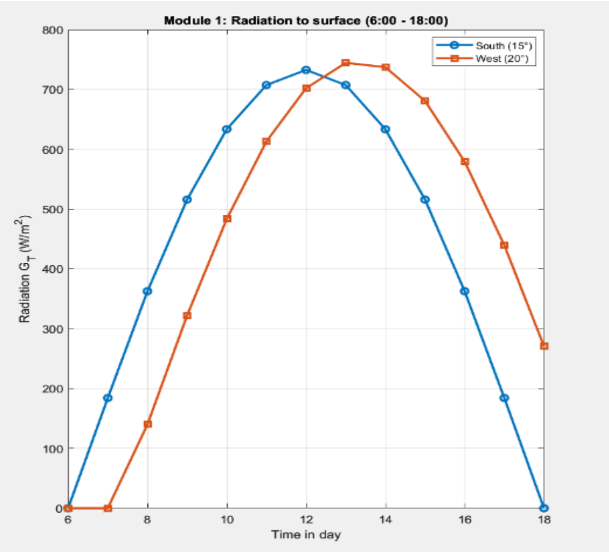
3.3. Assumptions and Limitations

- Neglecting Diffuse Radiation: The model only uses the clearness index k_T to calculate direct (beam) radiation, while ignoring diffuse and reflected radiation. In Ho Chi Minh City, the tropical weather has a lot of clouds, meaning diffuse radiation is actually very high. By ignoring this, the simulation might show an optimal tilt angle (β) that is steeper than it should be. In real life, the panels should be placed flatter to catch light from the whole sky.
- System Losses: The simulation ignores real-world factors like dust accumulation (soiling), shading, wiring resistance, and inverter efficiency. In reality, these parameters introduce a total system derate factor (typically 15–20%), which directly reduces the actual power output P_{out} relative to the ideal photovoltaic formulation.
- Simplified Temperature Effect: The calculation for cell temperature T_C only considers solar radiation and ambient temperature, while ignoring wind speed. Wind helps cool down the panels, which reduces the thermal degradation β_T and keeps the efficiency η higher.
- Averaged Weather and Fixed Tariff: The EVN electricity prices are assumed to be fixed for 1 year. Also, the model uses monthly average k_T data instead of real-

time hourly data. This smooths out sudden weather changes, like afternoon rain. Under the Time-of-Use (ToU) mechanism, the peak price is very high between 17:30 and 22:30, so using monthly averages might make the calculated revenue R for West-facing panels slightly higher than what they actually make.

4. RESULTS AND DISCUSSION

4.1. Physical Optimization (Maximum Yield)



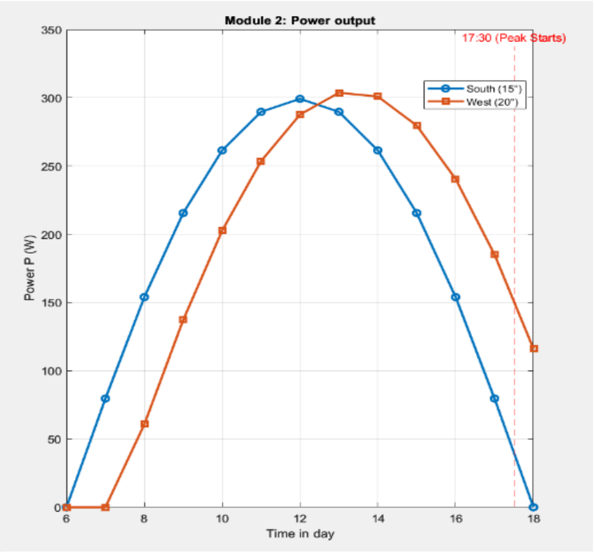


Figure 1. *Hourly Irradiance Profile* ***Figure 2.*** *Hourly Power Output Profile*

- Result:

  - South (15°): The amount of solar radiation and power output both begin to increase from 6:00 to 12:00. At 12:00, they reach their peak of 732.414 W\m^2 in radiation and 299.874 W in power generated. After that, they steadily fall.
  - West (20^0): The figures for the west begin to increase between 7:00 and 13:00. The highest solar irradiance and power wattage are 744.212 W\m^2 and 304.324 W, respectively. It is noteworthy that west-facing panels are exposed to more radiation and have higher power output during the peak hour, at 17:30, than its counterpart.

- Conclusion: South-facing at β=15° yields the highest annual energy of 788.59 kWh/year, confirming it as the physically optimal configuration.

4.2. Economic Optimization (Maximum Revenue)

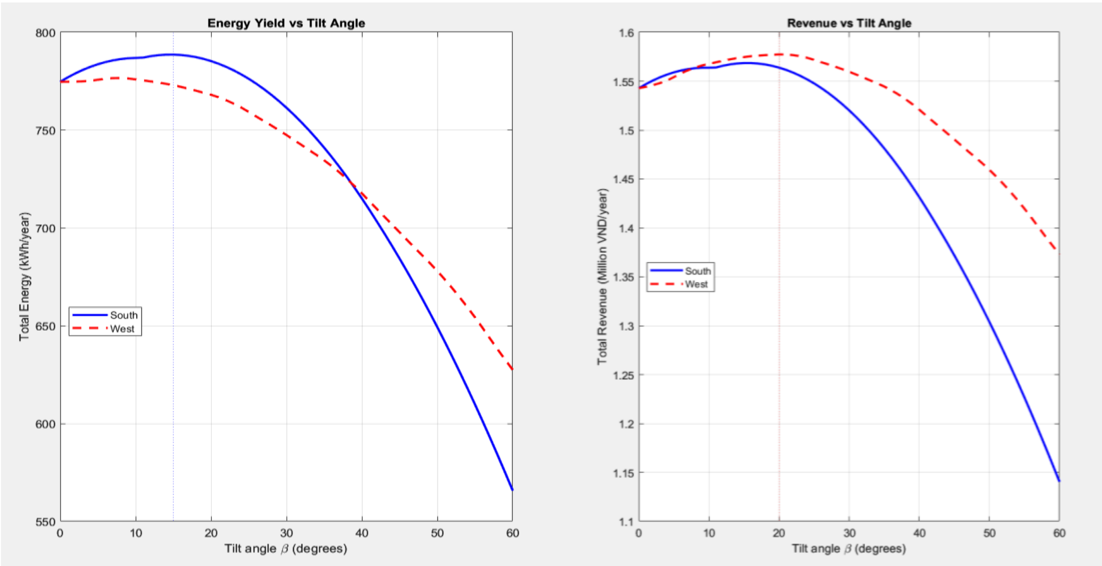


Figure 3. *Annual Energy Yield vs. Tilt Angle*

Figure 4. *Annual Revenue vs. Tilt Angle*

Result:

Energy output

- South orientation: Has a peak in annual energy yield of approximately 788.59 kWh\year. The optimal tilt angle is 15^o. The least amount of electricity produced is when the tilt angle is about 60^o.
- West orientation: Peaks when tilt angle is 8^o, achieves about 776.598 kWh\year. It is shown that west orientation power output reduces at a more gradual pace than southern orientation.

Revenue generation

- West orientation: Reaches the peak of 1,577,373 VND/year when tilting at 20^o. It then gradually reduces until it reaches its lowest point at 60^o.
- South orientation: Peaks at 1,568,540 VND\ year with roughly 15^o tilt angle. It is clear that the revenue for southern placement drops more sharply than its counterpart.

Conclusion:

- Positioning solar panels to the south generates more energy than to the west.
- For higher revenue, planting solar panels to the west is more ideal than to the south.

4.3. The Trade-off Analysis

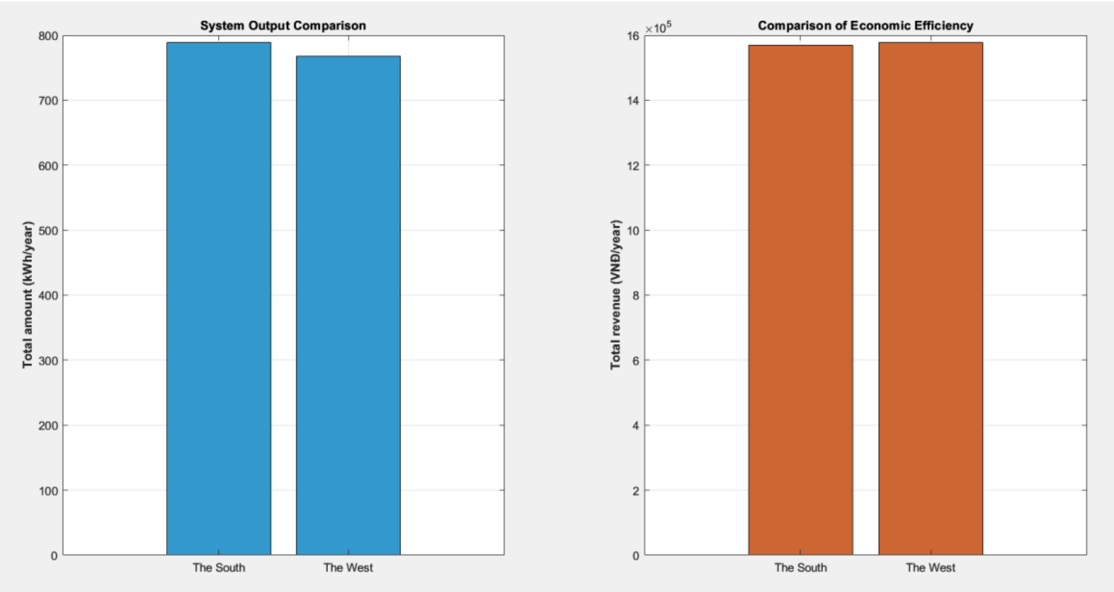


Figure 6. *Bar Chart: System Output and Revenue Comparison*

Result:

- System output comparison:
  - The south outperforms in yearly energy. The total electricity manufactured in the south is approximately 788.59 kWh\year.
  - The west has less total output than the south. Total power generated in the west is about 776.598 kWh\year.
- Comparison of economic efficiency: The west has a slight edge over the south in terms of annual revenue. This makes it more beneficial to revenue when planting solar panels facing the west.
- Conclusion: Despite having more annual power output, the revenue of south-facing solar panels still falls behind west oriented solar panels. This concludes that installing panels to the west offers more superior economic benefits than the south.

5. CONCLUSION

In conclusion, this study quantified the techno-economic trade-offs of fixed-tilt rooftop solar PV systems in Ho Chi Minh City by evaluating south- and west-facing configurations under EVN's industrial Time-of-Use tariff. The empirical results demonstrate that physical energy optimization and economic revenue maximization require different structural orientations. A south-facing panel at a 15° tilt maximizes annual energy yield, producing 788.59 kWh/year due to optimal solar geometry over the year. However, this setup concentrates generation during midday when electricity rates are lowest. Conversely, orienting the panel west at a 20° tilt alters the generation profile, sustaining a higher power output into the late afternoon. This shift allows the system to exploit EVN's peak pricing window (17:30–22:30), achieving a maximum annual revenue of 1,577,373 VND. Economically, the west-facing setup yields a 0.56% increase in financial returns at the expense of a marginal 2.60% loss in total energy output compared to the south-facing benchmark. Therefore, for grid-tied urban PV systems operating under time-differentiated tariffs without battery storage, optimizing for generation timing via a westward orientation offers superior financial efficiency over traditional maximum-yield configurations.

6. REFERENCES

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APPENDIX

%% STEP 1: CALCULATE THE ANGLE OF THE SUN AND THE ANGLE OF INCIDENCE

% 1. Declare geographical parameters Ho Chi Minh City

phi = 10.77; % Latitude (degrees)

% Declare the surface parameters of the solar panel

beta = 15;   % Tilt angle of the solar panel (degrees) - Will be changed in the outermost loop (0 - 60 degrees)

gamma = 0;   % Azimuth angle (degrees) - 0 degrees is South, 90 degrees is West

%% STEP 2: CALCULATE RADIATION ON THE INCLINED SURFACE (G_T)

% --- DATA DECLARATION ---

% k_T data for 12 months

% Month: 1   2   3   4   5   6   7   8   9   10   11   12

kT_array = [0.55, 0.60, 0.62, 0.55, 0.48, 0.49, 0.43, 0.50, 0.43, 0.47, 0.52, 0.50]; % this is january to december

% Array of the number of days in each month to convert day 'n' (1-365) to month

days_in_month = [31, 28, 31, 30, 31, 30, 31, 31, 30, 31, 30, 31];

cumulative_days = cumsum(days_in_month);   % [31, 59, 90, 120, ...]

G_sc = 1367;   % Solar constant G_sc (W/m^2)

% ==

% STEP 3 & 4: CALCULATE TEMPERATURE (Tc), EFFICIENCY (eta) AND POWER (P)

% ==

% --- DATA DECLARATION ---

% Solar panel specifications Canadian Solar HiKu CS3W-450MS

NOCT = 42;   % Nominal operating temperature (C degree)

eta_ref = 0.204;   % Reference performance (20.4%)

beta_T = 0.0035;   % Temperature coefficient (0.35%/C). Convert to a positive number because the formula already has a minus sign

T_ref = 25;   % Standard temperature STC (C degree)

A = 2.21;   % Solar panel area (m2)

```
% Average monthly ambient temperature (Ta) in HCMC
Ta_array = [26.54, 28.14, 29.99, 31.10, 29.93, 27.66, 26.90, 26.91, 26.82, 26.70, 26.70, 26.00];
% 2. Date and time loop
for n = 1:365 % The cycle of days in a year (1 to 365)

  %% STEP 1
  % Declination angle - delta
  % Using the standard Cooper formula in solar geometry
  delta = 23.45 * sind((360 / 365) * (284 + n));



  % Calculate the G_T radiation incident on the inclined solar panel surface
  % - theta: angle of incidence from Step 1 (in degrees)
  % - max(cosd(theta), 0): The max() function is used to eliminate negative radiation cases
  % (when cos(theta) < 0, it means the sun has not risen, has set, or is behind the solar panel)
  %% STEP 2
  % --- START CALCULATION (Place inside the 'n' and 'h' time loop) ---
  month_idx = find(n <= cumulative_days, 1); % Determine the current month (m) based on the day of the year (n)

  Ta = Ta_array(month_idx);   % Take the ambient temperature (Ta) for the current month.

  kT = kT_array(month_idx);  % Get the k_T value for the corresponding month

  % Calculate the extra-atmospheric solar radiation (G0) on day n
  G0 = G_sc * (1 + 0.033 * cosd(360 * n / 365));    % This is the radiation perpendicular to the sun's rays outside the atmosphere
  for h = 6:18 % Time loop (6 AM to 6 PM)
    %% STEP 1
    % Hour angle - omega
```

omega = 15 * (h - 12); % The sun is at its highest point at noon (h = 12), corresponding to omega = 0

% Zenith angle - theta_z

cos_theta_z = sind(phi) * sind(delta) + cosd(phi) * cosd(delta) * cosd(omega);

% Limit the value to the range [-1, 1] to avoid domain errors when calculating acosd due to rounding errors

cos_theta_z = max(min(cos_theta_z, 1), -1);

theta_z = acosd(cos_theta_z);

% Angle of incidence - theta - STEP 1

% General formulas for the beta inclined surface and gamma azimuth

```
cos_theta = sind(delta) * sind(phi) * cosd(beta) ...
    - sind(delta) * cosd(phi) * sind(beta) * cosd(gamma) ...
    + cosd(delta) * cosd(phi) * cosd(beta) * cosd(omega) ...
    + cosd(delta) * sind(phi) * sind(beta) * cosd(gamma) * cosd(omega) ...
    + cosd(delta) * sind(beta) * sind(gamma) * sind(omega);

cos_theta = max(min(cos_theta, 1), -1);
theta = acosd(cos_theta);
% STEP 1:
%% STEP 2: Radiation on an inclined surface G_T
GT = G0 * kT * max(cosd(theta), 0); % max(..., 0): eliminate the case of the sun behind the solar panel.

%% STEP 3: Temperature of battery (Tc) & efficiency (eta)
if GT > 0
  % NOCT: temperature increase according to the radiation
  Tc  = Ta + ((NOCT - 20) / 800) * GT;
  eta = eta_ref * (1 - beta_T * (Tc - T_ref));
else
  % No radiation → No electricity
  Tc  = Ta;
  eta = 0;
```

```
    end
    %% STEP 4: Power & Energy
    P = GT * A * eta;   % Power (W)
    E = P * 1;          % Energy per hour (Wh)

    %% PRINT
    % Display test results for day 1, hour 12 for verification
    if n == 1 && h == 12
      fprintf('=== STEP 1 ===\n')
      fprintf('--- Result on day %d, time %dh ---\n', n, h);
      fprintf('Latitude (phi): %.2f°\n', phi);
      fprintf('Angle of deviation (delta): %.2f°\n', delta);
      fprintf('Hour angle (omega): %.2f°\n', omega);
      fprintf('Zenith angle (theta_z): %.2f°\n', theta_z);
      fprintf('Incidence (theta): %.2f°\n\n', theta);
      fprintf('=== STEP 2 ===\n');
      fprintf('--- Result on day %d, time %dh ---\n', n, h);
      fprintf('Month: %d (kT = %.2f)\n', month_idx, kT);
      fprintf('Extraterrestrial radiation (G0): %.2f W/m2\n', G0);
      fprintf('Radiation from solar panels (GT): %.2f W/m2\n\n', GT);
      fprintf('=== STEP 3 ===\n');
      fprintf('--- Result on day %d, time %dh ---\n', n, h);
      fprintf('Ambient temperature (Ta): %.2f °C\n', Ta);
      fprintf('Panel cell temperature (Tc): %.2f °C\n', Tc);
      fprintf('Actual Efficiency (eta): %.2f %%\n', eta * 100); % Multiply by 100 to
display as a percentage

      fprintf('\n=== STEP 4 ===\n');
      fprintf('Output power (P): %.2f W\n', P);
      fprintf('Electricity generated (E): %.2f Wh\n\n', E);
    end
  end
end
```

```
%% STEP 6: BETA LOOP AND OPTIMAL POINT
% 1. Define the beta tilt angle range from 0 to 60 degrees (1-degree increments)
beta_array = 0:1:60;
num_betas = length(beta_array);
% 2. Initialize arrays to store results for the South (gamma = 0) and West (gamma = 90)
directions.
Energy_South = zeros(1, num_betas);
Revenue_South = zeros(1, num_betas);
Energy_West = zeros(1, num_betas);
Revenue_West = zeros(1, num_betas);
% Loop 1: Consider 2 directions (1: South, 2: West)
for dir = 1:2
  if dir == 1
     gamma = 0; % South
  else
     gamma = 90; % West
  end

  % Loop 2: Run the beta angle from 0 to 60 degrees
  for b_idx = 1:num_betas
     beta = beta_array(b_idx);

     % Reset the variable for the current (step 5)
     Total_Energy = 0;
     Total_Revenue = 0;

     % Loop 3: Day in the year------------------
     for n = 1:365
        % Determine the month and parameters according to the day n
        month_idx = find(n <= cumulative_days, 1);
        kT = kT_array(month_idx);      % line 29
        Ta = Ta_array(month_idx);      % line 59
```

```
delta = 23.45 * sind((360 / 365) * (284 + n));  % line 83
G0 = G_sc * (1 + 0.033 * cosd(360 * n / 365));  % line 33
% Loop 4: Hours in the Day-------------------
for h = 6:18
   % step 1. Hour angle and incident angle - line 89 to 106
   omega = 15 * (h - 12);
   cos_theta = sind(delta) * sind(phi) * cosd(beta) ...
           - sind(delta) * cosd(phi) * sind(beta) * cosd(gamma) ...
           + cosd(delta) * cosd(phi) * cosd(beta) * cosd(omega) ...
           + cosd(delta) * sind(phi) * sind(beta) * cosd(gamma) * cosd(omega) ...
           + cosd(delta) * sind(beta) * sind(gamma) * sind(omega);
   cos_theta = max(min(cos_theta, 1), -1);
   theta = acosd(cos_theta);
   % step 2. Radiation on an inclined surface
   GT = G0 * kT * max(cosd(theta), 0);
   % step 3. Temperature and efficiency
   if GT > 0
      Tc = Ta + ((NOCT - 20) / 800) * GT;
      eta = eta_ref * (1 - beta_T * (Tc - T_ref));
   else
      Tc = Ta;
      eta = 0;
   end
   % step 4. Power
   P = GT * A * eta;      % line 75
   E = P * 1;         % line 76
   % step 5. Revenue on hour
   % In 2026, January 1st will be a Thursday -> offset = 3 (0=CN, 1=T2, 2=T3, 3=T4, 4=T5, 5=T6, 6=T7)
   day_of_week = mod(n + 3, 7);
   % Determine electricity prices according to EVN's tariff schedule
   if h == 18 && day_of_week ~= 0      % 6 PM (represented by 5:30 PM - 6:30 PM) from Monday to Saturday: PEAK HOURS
```

```
                price = 3640;
            elseif h >= 6 && h <= 17            % 6 AM to 5 PM every day of the week: NORMAL HOURS
                price = 1987;
            elseif h == 18 && day_of_week == 0  % 6 PM on Sundays only: NORMAL
                price = 1987;
            else                                % Remaining hours (if any): OFF-PEAK HOURS
                price = 1300;
            end
            Revenue_h = (E / 1000) * price;
            % 6. Total
            Total_Energy = Total_Energy + E;
            Total_Revenue = Total_Revenue + Revenue_h;
        end % End loop hour h
    end % End loop day n

    % Store the results for one year in the array at the corresponding b_idx position.
    if dir == 1
        Energy_South(b_idx) = Total_Energy / 1000; % Change to kWh
        Revenue_South(b_idx) = Total_Revenue;
    else
        Energy_West(b_idx) = Total_Energy / 1000; % Change to kWh
        Revenue_West(b_idx) = Total_Revenue;
    end

  end % End loop b_idx
end % End loop dir
% =========================================================================
% FIND THE OPTIMAL POINT USING THE MAX() FUNCTION AND PRINT THE RESULT TO THE ABSTRACT
% =========================================================================
```

```
% 1. Physical Optimization (Maximum amount - South)
[max_energy_S, idx_energy_S] = max(Energy_South);
opt_beta_energy = beta_array(idx_energy_S);
% 2. Economic Optimization (Maximum revenue - West)
[max_rev_W, idx_rev_W] = max(Revenue_West);
opt_beta_rev = beta_array(idx_rev_W);
% 3. Print the result of step 6
fprintf('\n=== STEP 6 ===\n');
fprintf('Physical Optimization (South): \nAngle of inclination beta = %d°, \nOutput = %.2f kWh/year\n', opt_beta_energy, max_energy_S);
fprintf('Economic Optimization (West): \nAngle of inclination beta = %d°, \nRevenue = %s VND/year\n', opt_beta_rev, num2str(max_rev_W, '%10.0f'));
%% DRAW FIGURES 2 & 3: Electricity and Revenue according to beta
figure('Name', 'Optimization Curves', 'Position', [100, 550, 900, 400]);
% Figure 2: Annual electricity consumption according to beta
subplot(1,2,1);
plot(beta_array, Energy_South, 'b-', 'LineWidth', 2, ...
   'DisplayName', 'South'); hold on;
plot(beta_array, Energy_West, 'r--', 'LineWidth', 2, ...
   'DisplayName', 'West');
xline(opt_beta_energy, 'b:', 'HandleVisibility', 'off');
xlabel('Tilt angle \beta (degrees)');
ylabel('Total Energy (kWh/year)');
title('Energy Yield vs Tilt Angle');
legend('Location', 'best'); grid on;
% Figure 3: Annual revenue by beta
subplot(1,2,2);
plot(beta_array, Revenue_South/1e6, 'b-', 'LineWidth', 2, ...
   'DisplayName', 'South'); hold on;
plot(beta_array, Revenue_West/1e6, 'r--', 'LineWidth', 2, ...
   'DisplayName', 'West');
xline(opt_beta_rev, 'r:', 'HandleVisibility', 'off');
xlabel('Tilt angle \beta (degrees)');
```

```
ylabel('Total Revenue (Million VND/year)');
title('Revenue vs Tilt Angle');
legend('Location', 'best'); grid on;
%% STEP 7. DRAW DIAGRAM FOR COMPARISON
figure('Name', 'Trade-off Analysis: Output and Revenue', 'Position', [100, 100, 900, 400]);
% Chart 3: Comparison of output
subplot(1,2,1);
% Compare the output of the South direction (at the optimal power output angle) with that of the West direction (at the optimal economic angle)
bar([Energy_South(idx_energy_S), Energy_West(idx_rev_W)], 'FaceColor', [0.2 0.6 0.8]);
set(gca, 'XTickLabel', {'The South', 'The West'});
ylabel('Total amount (kWh/year)', 'FontWeight', 'bold');
title('System Output Comparison');
grid on;
% Chart 4: Revenue Comparison
subplot(1,2,2);
% Compare the revenue of the South direction (at the optimal power output angle) with the revenue of the West direction (at the optimal economic angle)
bar([Revenue_South(idx_energy_S), Revenue_West(idx_rev_W)], 'FaceColor', [0.8 0.4 0.2]);
set(gca, 'XTickLabel', {'The South', 'The West'});
ylabel('Total revenue (VNĐ/year)', 'FontWeight', 'bold');
title('Comparison of Economic Efficiency');
grid on;
%% DRAW DIAGRAM 1 (RADIATION) & 2 (POWER) BY EACH HOUR
%select_day = 'Select day: ';
%n_plot = str2double(input(select_day, 's'));
n_plot = 100;
month_idx_plot = find(n_plot <= cumulative_days, 1);
kT_plot = kT_array(month_idx_plot);
Ta_plot = Ta_array(month_idx_plot);
% Eccentric angle (delta) and radiation in atmosphere (G0) until now - eccentricity and radiation in the atmosphere
```

```
delta_plot = 23.45 * sind((360 / 365) * (284 + n_plot));
G0_plot = G_sc * (1 + 0.033 * cosd(360 * n_plot / 365));
% Time array from 6:00 to 18:00
hours = 6:18;
GT_South = zeros(1, length(hours)); P_South = zeros(1, length(hours));
GT_West = zeros(1, length(hours));  P_West = zeros(1, length(hours));
% Optimal angle - The optimal angle is determined from the code execution results
beta_S = opt_beta_energy;   % From physics optimization results
beta_W = opt_beta_rev;      % From economic optimization results
gamma_S = 0;
gamma_W = 90;
% Loop for radiation calculation (GT) & power (P) for each hour
for i = 1:length(hours)
  h_plot = hours(i);
  omega_plot = 15 * (h_plot - 12);

  % --- 1. CALCULATE FOR THE SOUTH ---
  cos_theta_S = sind(delta_plot)*sind(phi)*cosd(beta_S) -
sind(delta_plot)*cosd(phi)*sind(beta_S)*cosd(gamma_S) +
cosd(delta_plot)*cosd(phi)*cosd(beta_S)*cosd(omega_plot) +
cosd(delta_plot)*sind(phi)*sind(beta_S)*cosd(gamma_S)*cosd(omega_plot) +
cosd(delta_plot)*sind(beta_S)*sind(gamma_S)*sind(omega_plot);
  GT_South(i) = G0_plot * kT_plot * max(cos_theta_S, 0);

  if GT_South(i) > 0
     Tc_S = Ta_plot + ((NOCT - 20) / 800) * GT_South(i);
     eta_S = eta_ref * (1 - beta_T * (Tc_S - T_ref));
  else
     eta_S = 0;
  end
  P_South(i) = GT_South(i) * A * eta_S;

  % --- 2. CALCULATE FOR THE WEST ---
```

```
  cos_theta_W = sind(delta_plot)*sind(phi)*cosd(beta_W) - sind(delta_plot)*cosd(phi)*sind(beta_W)*cosd(gamma_W) + cosd(delta_plot)*cosd(phi)*cosd(beta_W)*cosd(omega_plot) + cosd(delta_plot)*sind(phi)*sind(beta_W)*cosd(gamma_W)*cosd(omega_plot) + cosd(delta_plot)*sind(beta_W)*sind(gamma_W)*sind(omega_plot);
  GT_West(i) = G0_plot * kT_plot * max(cos_theta_W, 0);


  if GT_West(i) > 0
     Tc_W = Ta_plot + ((NOCT - 20) / 800) * GT_West(i);
     eta_W = eta_ref * (1 - beta_T * (Tc_W - T_ref));
  else
     eta_W = 0;
  end
  P_West(i) = GT_West(i) * A * eta_W;
end
% Diagram code
figure('Name', 'Module 1 & 2: Radiation and power according to each hour', 'Position', [150, 150, 1000, 450]);
% Chart 1 (Radiation - Module 1)
subplot(1,2,1);
plot(hours, GT_South, '-o', 'LineWidth', 2, ...
  'DisplayName', sprintf('South (%d°)', beta_S)); hold on;
plot(hours, GT_West, '-s', 'LineWidth', 2, ...
  'DisplayName', sprintf('West (%d°)', beta_W));
title('Module 1: Radiation to surface (6:00 - 18:00)');
xlabel('Time in day');
ylabel('Radiation G_T (W/m^2)');
legend('Location', 'best');
grid on;
xlim([6 18]);
% Chart 2 (Power - Module 2)
subplot(1,2,2);
plot(hours, P_South, '-o', 'LineWidth', 2, ...
  'DisplayName', sprintf('South (%d°)', beta_S));
```

```
hold on;
plot(hours, P_West, '-s', 'LineWidth', 2, ...
  'DisplayName', sprintf('West (%d°)', beta_W));
xline(17.5, '--r', '17:30 (Peak Starts)', ...
  'LabelOrientation', 'horizontal', ...
  'LabelHorizontalAlignment', 'center', ...
  'HandleVisibility', 'off');
title('Module 2: Power output');
xlabel('Time in day');
ylabel('Power P (W)');
legend('Location', 'best'); grid on; xlim([6 18]);
```